Portable STM32-Based Cardiac-Fall Fusion Monitor Using On-Device TinyML ECG Analysis

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Preliminary Study (500 words)

Heart disease remains one of the leading causes of death worldwide, and a dangerous heart event becomes far more life-threatening when it happens alongside a fall or loss of consciousness — a risk that is especially high for elderly people or anyone living alone. Outside a hospital, people currently have limited options: periodic checkups that only capture a snapshot in time, wearable devices that record data for a doctor to review later, or basic fitness trackers that measure heart rate but cannot interpret whether the heart's rhythm is actually abnormal.

A review of recent published work shows that the closest existing systems each solve only part of this problem. Some systems detect heart rhythm problems using AI running directly on a small microcontroller, but they are tested only on old, pre-recorded heart signal data rather than a live sensor reading an actual heartbeat, and can only flag something as abnormal without classifying the type of problem. Other systems combine heart monitoring with fall detection, but they track only basic heart rate rather than the true electrical signal of the heart, with intelligence running on a remote server rather than on the device itself. Systems that do read the heart's real electrical signal and analyze its shape typically depend on a separate connected computer and do not track falls at all. No system found in this review combines live heart-signal reading, on-device AI analysis, and fall detection together, such that a rhythm problem and a fall occurring at the same time are recognized as a single, higher-priority emergency rather than two separate, unrelated alerts.

This project addresses that specific gap. We are building a small, battery-powered device that reads a person's heartbeat in real time through an ECG sensor, processes and analyzes the signal using a lightweight AI model (TinyML) running entirely on an STM32 microcontroller — with no internet connection or separate computer required — and combines this with a fall-detection sensor. If the device detects a concerning heart rhythm and a fall occurring close together in time, it raises the alert to a higher priority than either signal alone would trigger. To keep the project realistic within the timeline, the system will first flag heart rhythm as normal or abnormal, with classification into more specific rhythm categories pursued as a further goal if time permits, while keeping the entire detection and decision process on the device itself rather than dependent on cloud processing or a network connection.

Objectives of the Project Work

- Design a portable, battery-powered device that continuously reads a person's ECG in real time using chest electrodes.
- Process the raw ECG signal on an STM32 microcontroller using digital filtering, and calculate heart rate and detect rhythm irregularities.
- Develop and deploy a lightweight on-device AI (TinyML) model to classify heart rhythm as normal or abnormal, with finer classification as a stretch goal.
- Integrate a motion sensor (MPU6050) to detect fall events in real time from the body's motion signal.
- Design a decision-fusion mechanism that raises a higher-priority alert when an abnormal rhythm and a fall are detected together.
- Display real-time status on an on-device screen and trigger a local alarm, with no dependency on internet or a paired computer for detection.
- Evaluate the system's own measured accuracy, response time, and power consumption against relevant published work.

Component and Software Required to Carry Out the Project

Component/Software	Purpose
AD8232 ECG Sensor Module	Acquires the heart's electrical signal (ECG) through chest electrodes
ECG Electrodes (disposable)	Skin-surface contact for ECG signal pickup
STM32 Microcontroller	Main processing unit for signal processing and on-device AI inference
MPU6050 Accelerometer/Gyroscope	Detects fall events and body movement
0.96" OLED Display	Displays heart rate, rhythm status, and alerts in real time
Buzzer & LED	Local audible/visual emergency alarm
Li-ion Battery + TP4056 Charging Module	Portable rechargeable power supply

Component/Software	Purpose
Push Button(s)	Manual reset / basic user interface
STM32CubeIDE / STM32CubeMX	Firmware development and peripheral configuration
Python (TensorFlow/Keras) + TensorFlow Lite for Microcontrollers	TinyML model training, quantization, and on-device deployment
MIT-BIH Arrhythmia Database (PhysioNet)	Public ECG dataset used for training and evaluating the AI model

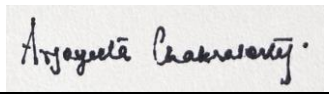
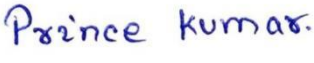
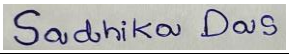
Cost Estimation of the Project

Item	Approx. Cost (₹)
AD8232 ECG Module	600
ECG Electrodes (pack)	150
MPU6050 Accelerometer/Gyroscope	180
STM32 Microcontroller Board	1,800
0.96" OLED Display	350
Buzzer, LED, Push Button	100
Li-ion Battery (18650)	500
TP4056 Charging Module	80
Wires, PCB/breadboard, enclosure	600
Contingency / spare components (~15%)	650
Total (approx.)	5,010

Note: All costs are approximate and should be confirmed with current vendor pricing before procurement.

Work Distribution Among the Project Group Members

Member	Responsibility
Sadhika Das	Signal Acquisition & Processing: AD8232-STM32 ADC interfacing; digital filtering; R-peak/heart-rate calculation; MPU6050 fall-detection integration; and fusion/escalation logic combining rhythm and fall status.
Arjoyeeta Chakraborty	TinyML Intelligence Layer: MIT-BIH dataset preprocessing; model design, training and validation; INT8 quantization and TensorFlow Lite Micro conversion; on-device deployment and accuracy documentation.
Prince Kumar	Signal Acquisition & Processing for ECG monitoring; Hardware & Integration: component sourcing, wiring and board bring-up (OLED, buzzer, battery, charging); enclosure assembly; integrating all modules into one device; end-to-end testing; and leading documentation/reporting.

Name	Signature	Date
Arjoyeeta Chakraborty (2304014)		22.07.26
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Signature of Supervisor